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SUPPLEMENTARY INFORMATION

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Nutrient recycling facilitates long-term stability of marine microbial phototroph-heterotroph interactions

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Supplementary Note 1

R. pomeroyi DSS-3 ALSO SUSTAINED GROWTH OF OTHER AXENIC *Synechococcus* STRAINS

We tested co-culturing the heterotroph *R. pomeroyi* DSS-3 with other available axenic and non-axenic *Synechococcus* strains (Supplementary Figure 1). Again, the axenic strains *Synechococcus* spp. WH8102 and WH7805 died after 80-100 and 20-40 days, respectively, but not when they were co-inoculated with *R. pomeroyi* DSS-3. As expected, the non-axenic strains *Synechococcus* spp. CC9311 and BL107 (which contain natural co-occurring contaminant heterotrophs within the culture) showed no difference whether they were co-inoculated or not with *R. pomeroyi* DSS-3 (Supplementary Figure 1).

Supplementary Note 2

GROWTH LIMITATION OF THE HETEROTROPH IN *Synechococcus* CULTURES

The heterotrophs in co-culture with *Synechococcus* sp. WH7803 reached high cell densities despite the absence of any exogenous addition of an organic source of carbon or energy (~5×10^8 CFU ml^{-1} for *R. pomeroyi* and 2-3×10^9 CFU ml^{-1} for the *Tropicibacter*-like strain) (Figure 1d in the main text). However, the peak in cell abundance was only obtained after approximately 70 days indicating slow growth of the heterotroph (i.e. 267 and 234 h doubling times, respectively, when excluding growth within the first 3 days). These high cell densities were not observed in the absence of the phototroph (Supplementary Figure 2). Genomic[1] and experimental analyses show *R. pomeroyi* DSS-3 cannot grow using nitrate as the sole source of nitrogen and that it is an auxotroph for the vitamin biotin. Since ASW medium contains nitrate and no vitamin or organic carbon source, *R. pomeroyi* relies on the photosynthate produced by *Synechococcus* to supply these essential nutrients. The supply of a *Synechococcus* cell lysate (produced as a result of viral lysis or mechanical sonication) to *R. pomeroyi* cultures in ASW medium produced rapid initial growth (doubling times of 4.6 and 5.4 h, respectively) followed by entry into the stationary phase (cell density: 10^7 cells ml^{-1}; Supplementary Figure 2). On the other hand addition of 'live' *Synechococcus* cells to the *R. pomeroyi* culture supplied continuously leaked photosynthate that, in the long term, resulted in a 10-fold higher cell density of heterotrophic bacterial cells (10^8 cells ml^{-1}). Hence, *R*. *pomeroyi* clearly benefits from this mutualistic interaction through the acquisition of photosynthate as a source of carbon and energy. This beneficial effect is unsurprisingly higher if *Synechococcus* is alive and continuously producing and releasing organic matter.

Supplementary Note 3

PHYSICOCHEMICAL ASPECTS OF ASW CULTURES

Auxotrophic needs: All marine *Synechococcus* strains tested in this study have no known auxotrophic needs and, hence, were routinely grown in ASW, a medium containing only inorganic nutrients and no auxotrophic supplements (e.g. vitamins).

Reactive oxygen species and pH: To date, oxidative stress was suggested as a major driving force during mutualistic picocyanobacteria and heterotroph interactions[2,3]. However, genomic analysis[4] and experimental evidence (see proteomic data) indicates that *Synechococcus* sp. WH7803 encodes and produces all necessary mechanisms to deal with oxidative stress. Despite the fact that light irradiation of organic matter can generate ROS[5], such radicals are depleted from the growth media over time in both axenic and co-cultures in stationary phase (Supplementary Figure 3a) and the addition of catalase to axenic cultures did not increase the lifespan of the culture. Over time, ROS is depleted from the media even when utilising catalase/peroxidase deficient *Prochlorococcus* strains (Figure S1 in[6]). Hence, ROS is not responsible for the death of long-term phototrophic cultures. Axenic and co-cultures also showed no difference in pH (pH ~8.8 in late stationary phase).

Nutrient limitation: Additional nutrients were added to 30-day old cultures to assess whether the death of axenic cultures was due to nutrient depletion (Supplementary Figure 3b). Nutrient addition did not have any effect on axenic cultures and the extended survival of *Synechococcus* was only observed when the heterotroph was present. Indeed, we found that *Synechococcus* cell densities in ASW medium are limited by light (presumably due to self-shading) and not by depletion of any of the key macro- or micro-nutrients in the media (Supplementary Figure 3c).

Supplementary Note 4

CELL COUNT DETERMINATION

Ruegeria pomeroyi DSS-3

Cells counts used to monitor a time course of axenic *R. pomeroyi* cultures grown in sterile natural seawater (SW; Sigma) were performed via i) plate counts (colony forming units; CFU) on marine agar (Difco) plates; and ii) flow cytometry (BD Fortessa) after SYTOX Green (Thermo Fisher Scientific) staining following the manufacturer's instructions (Supplementary Figure 6a). SYTOX is a DNA-binding fluorescent dye and is used as a live/dead cell indicator, i.e. live cells with an active membrane potential will exhibit a lower fluorescence intensity since the dye will be actively pumped out of the cell.

Following SYTOX staining high and low intensity fluorescing populations were observed which were attributed to dead and live cells, respectively. CFUs correlated well with 'live' cells counted by flow cytometry and confirmed the use of CFU as a good *proxy* for counting *R. pomeroyi*.

Synechococcus

Due to the low plating efficiencies of marine *Synechococcus* on solid medium *Synechococcus* sp. WH7803 cell abundance was routinely monitored by flow cytometry (BD FACScan). Such an approach is facilitated by the natural orange (due to phycoerythrin) and red (due to chlorophyll *a*) autofluorescence of this genus. Axenic *Synechococcus* sp. WH7803 cultures were also stained with SYTOX in order to compare cell counts obtained using natural autofluorescence with viable cell counts. We monitored a time course of axenic *Synechococcus* sp. WH7803 incubated in sterile natural SW and compared stained and unstained samples using flow cytometry (Supplementary Figure 6b).

Autofluorescent cell counts compared well with those obtained from live SYTOX staining (Supplementary Figure 6b). Complete cell death was observed at day 35 by loss of both autofluorescent pigmentation and membrane potential.

SUPPLEMENTARY REFERENCES

1. Moran, M. A. *et al*. Genome sequence of *Silicibacter pomeroyi* reveals adaptations to the marine environment. *Nature* **432,** 910-913 (2004).
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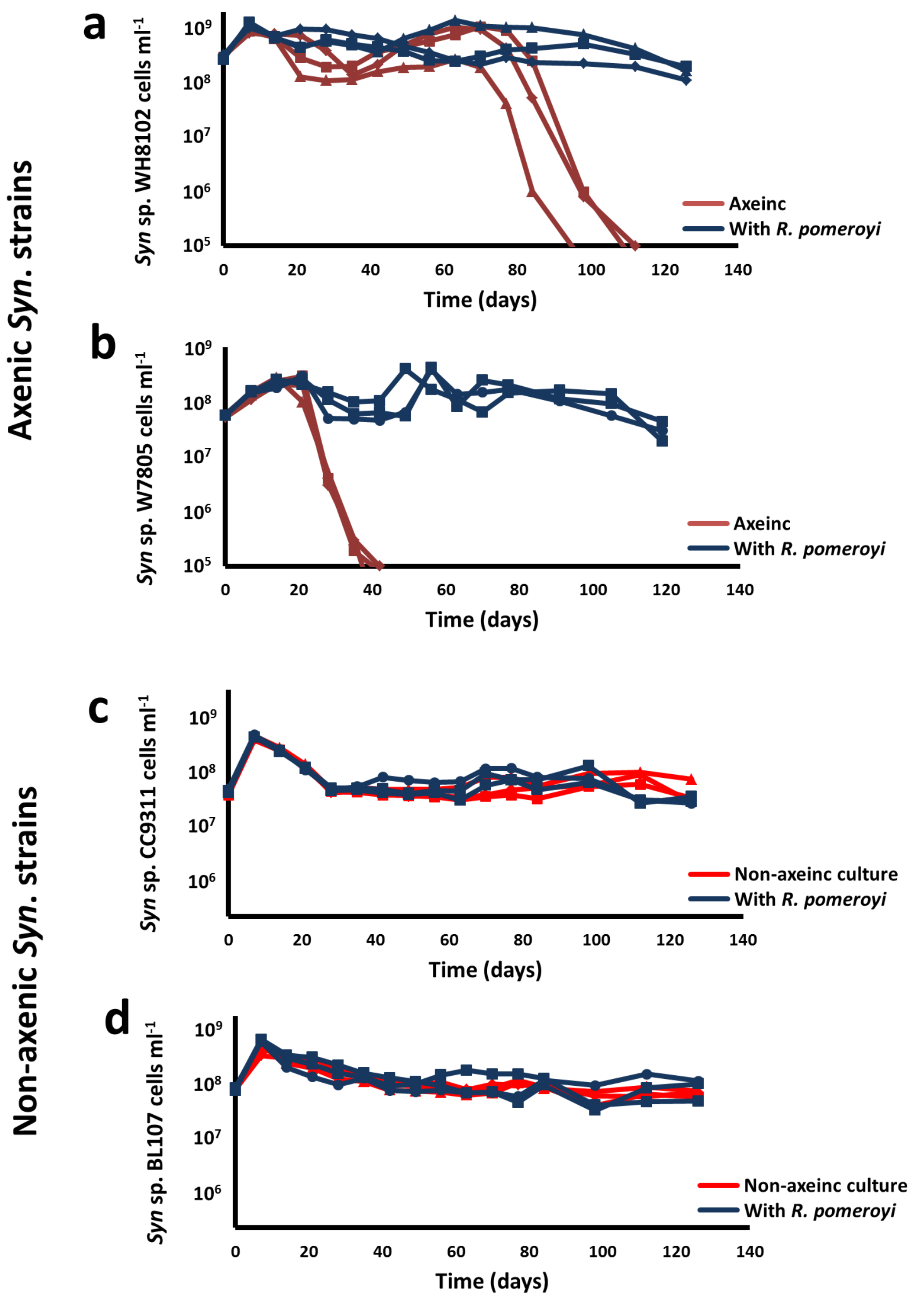
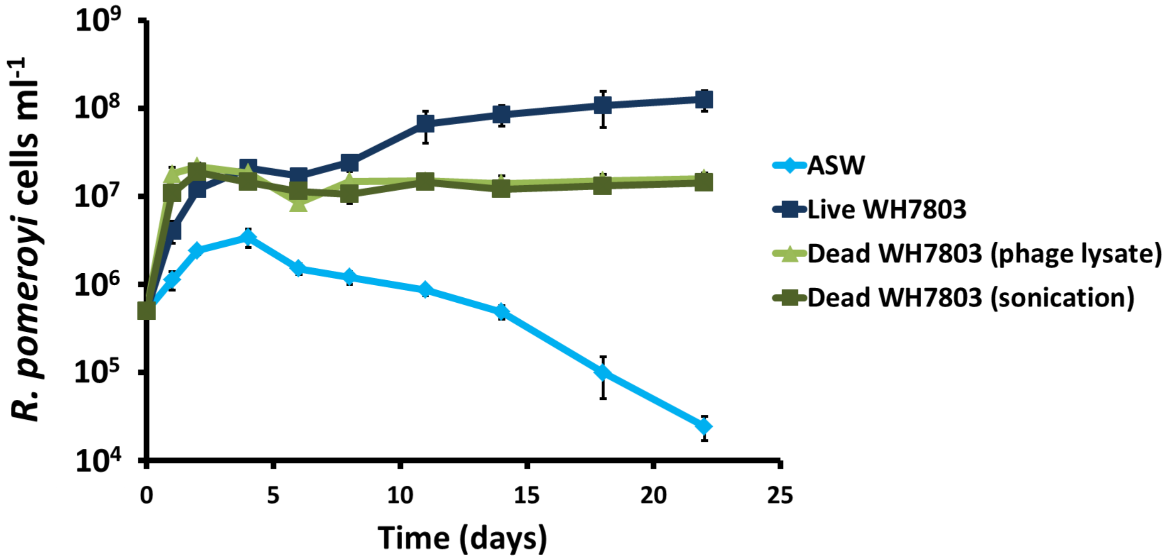
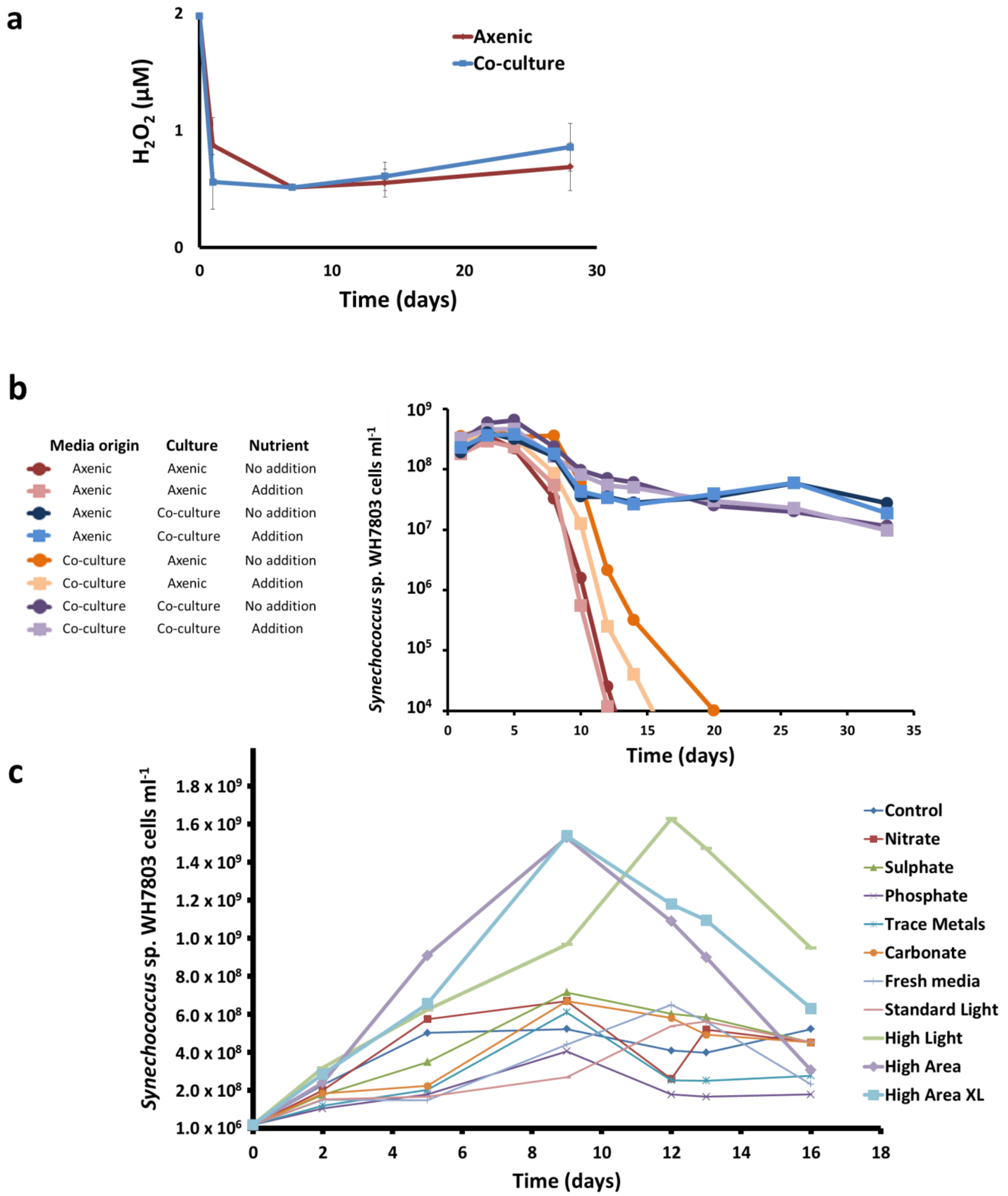
SUPPLEMENTARY FIGURES



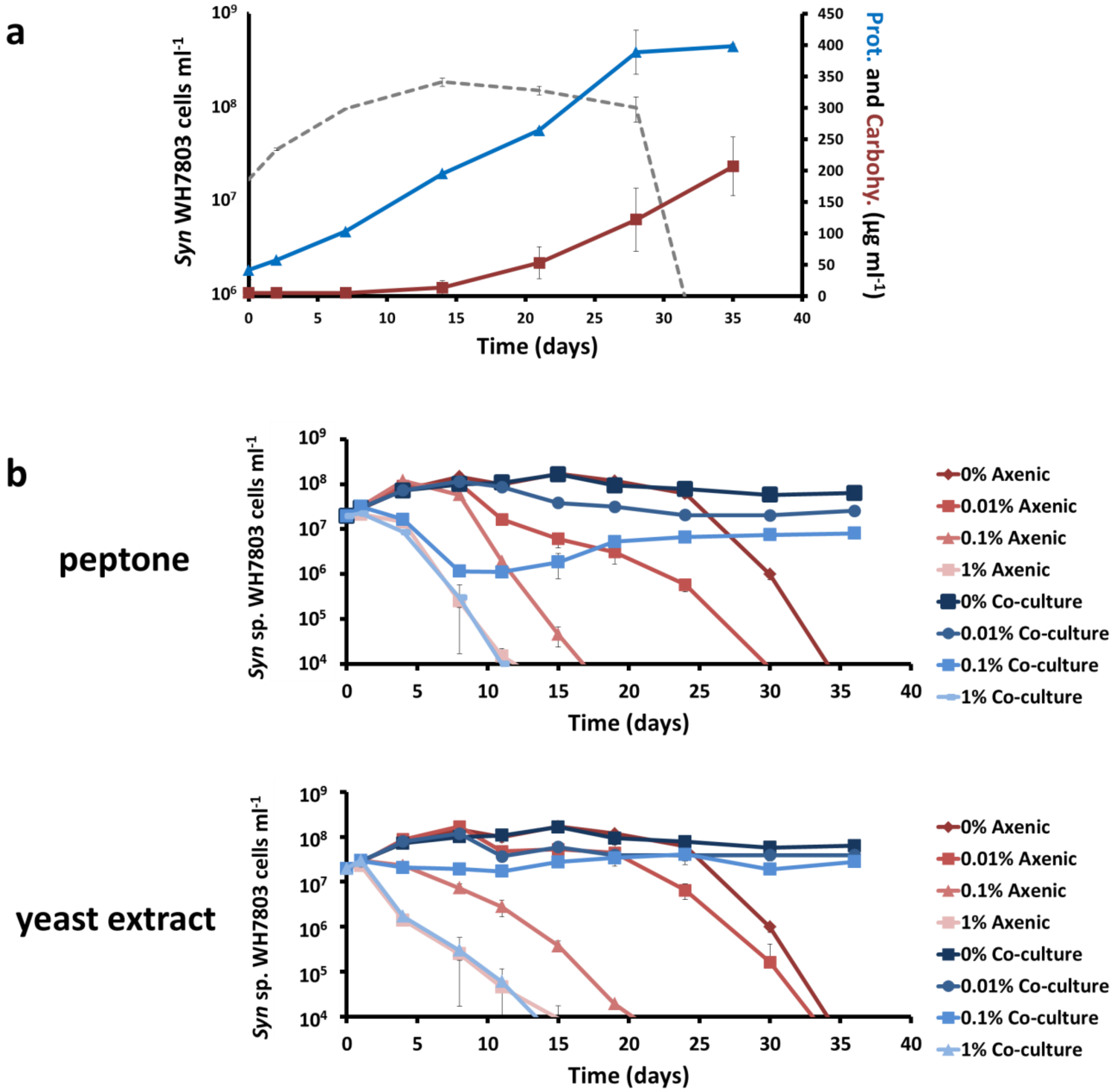
Supplementary Figure 1. Growth of (**a**) axenic *Synechococcus* sp. WH8102, (**b**) axenic *Synechococcus* sp. WH7805, (**c**) non-axenic *Synechococcus* sp. CC9311 and (**d**) non-axenic *Synechococcus* sp. BL107 in the presence/absence of *R. pomeroyi* DSS-3. *Synechococcus* cell counts were determined by flow cytometry. Three culture replicates (n=3) of each condition are represented.



Supplementary Figure 2. Growth curves of *R. pomeroyi* DSS-3 when grown in ASW and in ASW with live and dead *Synechococcus* cells. An equal number of live and dead cells were added at the start of the experiment ($3x10^8$ cells ml^{-1}). The average value from triplicate cultures (n=3) is shown (error bars show standard deviation).

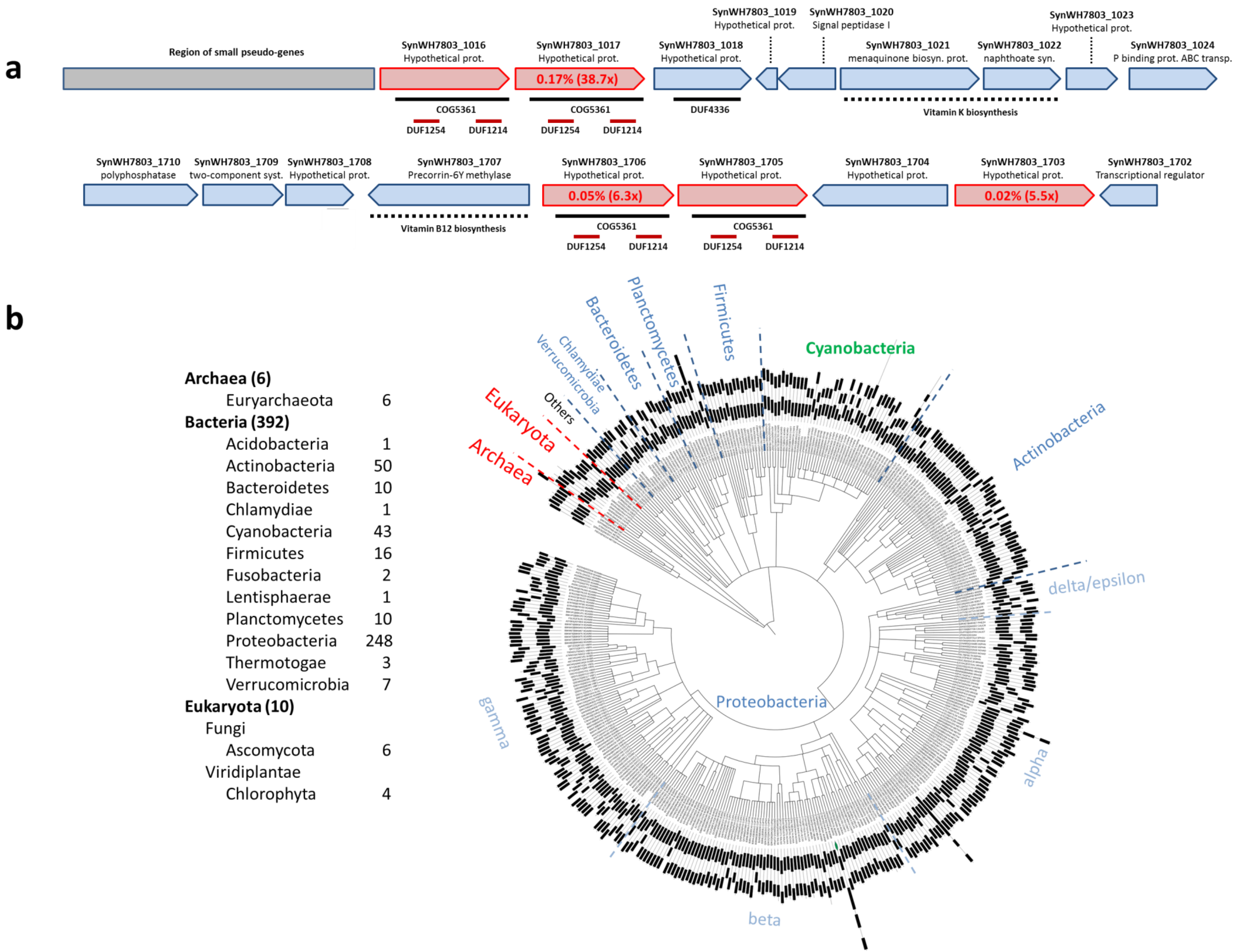


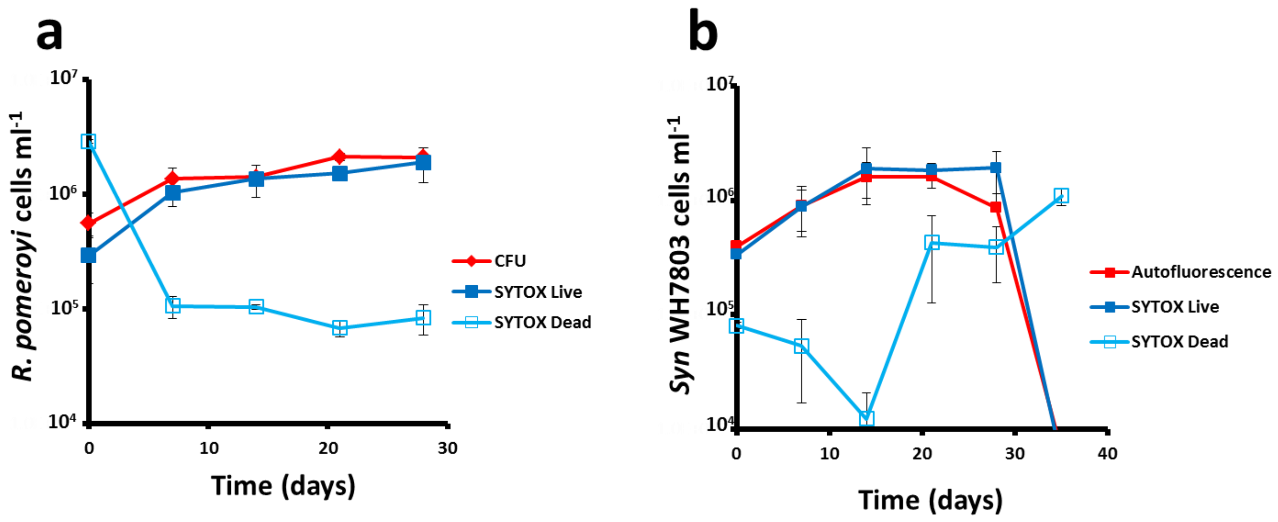
Supplementary Figure 3. (**a**) The concentration of hydrogen peroxide in ASW grown *Synechococcus* sp. WH7803 cultures in the presence and absence of *R. pomeroyi* DSS-3. (**b**) Growth curves of *Synechococcus* sp. WH7803 cultures in the presence and absence of *R. pomeroyi* DSS-3 inoculated in filter-sterilised media from 24-day old axenic and non-axenic cultures. An extra 50% of the normal nutrients found in ASW was added where indicated. (**c**) Growth limitation of *Synechococcus* sp. WH7803 in ASW. Each nutrient was increased by double the normal concentration in ASW. Sodium bicarbonate was added at a final concentration of 10 mM. The light intensity was doubled to 20 µmol photons m^{-2} s^{-1} under the 'high light' condition. Larger flasks were used to obtain a higher area to volume ratio. The average value from triplicate cultures (n=3) is shown in all panels (error bars show standard deviation).



Supplementary Figure 4. (**a**) Protein (blue) and carbohydrate (red) accumulation over time in axenic *Synechococcus* sp. WH7803 cultures grown in ASW. The growth of the cyanobacterium is shown (grey dashed line). (**b**) Growth curves of *Synechococcus* sp. WH7803 in ASW in the presence/absence of *R. pomeroyi* DSS-3 and after the addition of different concentrations of organic matter (i.e. peptone and yeast extract). The average value from triplicate cultures (n=3) is shown in all panels (error bars show standard deviation).

Supplementary Figure 5. *Synechococcus* sp. WH7803 proteins P1017 and P1706 up-regulated in the presence of *R. pomeroyi* DSS-3. (**a**) Genomic context and protein architecture. Both genes encoding P1017 and P1706 are located adjacent to another two CDS containing COG5361. The relative abundance of each protein and fold change (in brackets) in the presence of the heterotroph is indicated. (**b**) Proteins containing the same domain structure as P1017 are present in 210/1,133 microbial genomes using EMBL–SMART software (Table S3, Letunic et al. 2014). The outer ring of the phylogenetic tree generated by SMART depicts the protein architecture (DUF1254 and DUF1214) in all 408 proteins detected.





Supplementary Figure 6. Growth curves of *R. pomeroyi* DSS-3 (**a**) and *Synechococcus* sp. WH7803 (**b**) in natural seawater. *R. pomeroyi* was measured using colony forming units (CFU) and flow cytometry following staining with SYTOX. *Synechococcus* sp. WH7803 was measured by flow cytometry using autofluorescence and after staining the cells with SYTOX. The average value from triplicate cultures (n=3) is shown in all panels (error bars show standard deviation).

SUPPLEMENTARY TABLES

Supplementary Table 1A: Non-redundant peptide list of the Artificial Sea Water (ASW) cell proteome obtained from the three axenic and three *R*. *pomeroyi* co-inoculated cultures of *Synechococcus* sp. WH7803.

Supplementary Table 1B: Polypeptide list of the Artificial Sea Water (ASW) cellular proteome of the three axenic and three *R*. *pomeroyi* co-inoculated cultures of *Synechococcus* sp. WH7803 (proteins validated with 2 or more different peptides)

Supplementary Table 2A: Protein categories and comparative proteomic analysis of *Synechococcus* sp WH7803 proteins detected in ASW axenic and *R*. *pomeroyi* co-cultures.

Supplementary Table 2B: Protein categories of *R*. *pomeroyi* proteins detected in ASW co-culture with *Synechococcus* sp. WH7803.

Supplementary Table 3: Copy number of genes containing COG5361 in 1,133 genomes in EMBL - SMART

Supplementary Table 4A: Non-redundant peptide list of the Sea Water (SW) cell proteome obtained from the three axenic and three *R*. *pomeroyi* co-inoculated cultures of *Synechococcus* sp. WH7803.

Supplementary Table 4B: Polypeptide list of the Sea Water (SW) cellular proteome of the three axenic and three *R*. *pomeroyi* co-inoculated cultures of *Synechococcus* sp. WH7803 (proteins validated with 2 or more different peptides)

Supplementary Table 5A: Protein categories and comparative proteomic analysis of *Synechococcus* sp. WH7803 proteins detected in SW axenic and *R*. *pomeroyi* co-cultures.

Supplementary Table 5B: Protein categories of *R*. *pomeroyi* proteins detected in SW co-culture with *Synechococcus* sp. WH7803.

Supplementary Table 6A: Non-redundant peptide list of the Sea Water (SW) cell proteome obtained from *R*. *pomeroyi* in the presence/absence of *Synechococcus* sp. WH7803.

Supplementary Table 6B: Polypeptide list of the Sea Water (SW) cellular proteome of *R. pomeroyi* in the presence/absence of *Synechococcus* sp. WH7803 (proteins validated with 2 or more different peptides).

Supplementary Table 7: Protein categories of *R*. *pomeroyi* proteins detected in SW cultures in the presence/absence of *Synechococcus* sp. WH7803.